

Netcat: The TCP/IP Swiss Army Knife

Here's an introduction to Netcat, the powerful computer networking utility that performs numerous network related tasks. It is designed to be a dependable back-end, which can be run independently or in conjunction with other programs and scripts.



If you are worried about whether Netcat is software or a knife, as the headline states, let me clear your doubts — it is an open source tool that has a lot of functionalities, just like the Swiss army knife.

Netcat is an open source UNIX utility written in C. The name Netcat is derived from 'Net' (for networking) and 'cat' for concatenating data to a file. So, Netcat can be described as a tool for performing network related tasks and storing the output of the operations into a file, if required. This tool has been developed by someone who uses the pseudonym Hobbits. The beauty of the tool is that it can be either used directly from the terminal or be controlled by a user written program. You can use Netcat for creating a simple chat server, to scan the network for open ports, and more. It has the ability to send data packets over the network using the TCP and UDP protocols; however, TCP is used by default.

Versions of Netcat

- Netcat-traditional: The original version of Netcat
- Netcat-OpenBSD: Developed Netcat with some more options
- Ncat: The Netcat version developed by the Nmap community

In this article, I will be using Netcat-OpenBSD.

Terms used

Listener: A computer in which Netcat is listening on a port.

Client: A computer that tries to connect and/or give

commands to another computer on which Netcat is running.

Installation

To install Netcat, run the following code:

```
sudo apt-get install netcat-openbsd
```

Technical details

There are two systems (VMs) set up on the network with the following IP addresses:

- Ubuntu-Mate: 192.168.56.102 (client)
- Ubuntu: 192.168.56.103 (listener)

Uses of Netcat

Creating a chat server: A chat server is a system used for the purpose of chatting. Using Netcat, creating a simple chat server for two people to chat is very easy. You just need to type the following commands.

For the listener, type:

```
nc -lp <PORT>
```

For the client, type:

```
nc <IP ADDRESS> <PORT>
```

...where <IP ADDRESS> is the IP address of the